ANTHONY MARTINEZ

As a dedicated data scientist, I specialize in designing reproducible data analysis workflows that enhance transparency and reliability in scientific research. With experience in effectively communicating complex data patterns through insightful visualizations, I empower decision-makers with clear, actionable insights. Committed to continuous improvement in the field, I actively capture and share knowledge of data science best practices through training sessions and published resources, fostering a culture of collaboration and reproducibility within the data science community. My collaborative approach and technical expertise enable me to contribute significantly to projects, ensuring that scientific outputs are both robust and accessible.

EDUCATION

2019

M.S. Natural Resources

University of Idaho — Moscow, ID

- **Thesis**: Characterizing and ranking the importance of fire refugia in the northwestern US[1]

2017

B.S. Environmental Science and Resource Management, *cum laude*

University of Washington — Seattle, WA

- **Honors thesis**: Wildfire tree mortality detection and prediction using LiDAR and Landsat
- **Minor**: Quantitative science

RESEARCH EXPERIENCE

current | 2022

Data Scientist

Data Science Branch | U.S. Geological Survey — Loveland, CO

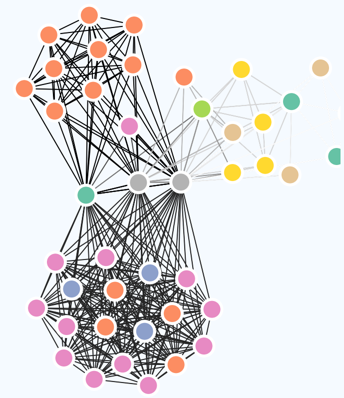
- Develop reproducible workflows for analyzing large, complex datasets for scientific research.
- Communicate complicated, multidimensional data patterns with innovative and engaging data visualizations.
- Spearheaded initiatives to capture and disseminate data science best practices through blog posts and targeted training sessions.

2019 | 2017

Research Assistant

Natural Resources and Society | University of Idaho — Moscow, ID

- Designed and executed natural resource management-focused and remote sensing/GIS-based research projects. Collected and analyzed data and communicated data collaboratively with other researchers.
- Designed participatory GIS survey. Developed multi-criteria decision analysis model.



View this CV online with links at https://amart90.github.io/cv/

CONTACT

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amart90
anthonymartinez90
usgs.gov/staff-profiles/anthony-martinez

LANGUAGE SKILLS

R
Python
HTML

Made with the R packages ***pagedown*** *and* ***datadrivencv***.

Last updated on 2025-01-24.

2017 | 2015

Research Assistant

Wildland Fire Sciences Lab | University of Washington — Seattle, WA

- Designed predictive model to estimate wildfire severity using remotely sensed forest structure metrics.

OTHER EXPERIENCE

2022 | 2019

Forester

Nez Perce-Clearwater National Forest | USDA Forest Service — Orofino, ID

- Managed 800,000 acres of forest by assessing current conditions and prescribing and implementing silvicultural treatments to achieve desired forest conditions.

2013 | 2008

Hospital Corpsman

U.S. Navy — Naval Hospital Guantanamo Bay, Cuba

- Purchasing agent and medic for a small, isolated military hospital. Operated with little direct supervision in a high-stress, highly team-oriented environment.

My diverse work experience has equipped me with valuable skills and perspectives that have shaped my approach as a data scientist.

DATA SCIENCE WRITING AND TRAINING

2024

Reproducible Data Science in R: Writing functions that work for you[2]

USGS Water Data for the Nation blog

ED Hinman and **AJ Martinez** | Learn the ropes building your own functions in R using water data examples.

2024

Reproducible Data Science in R: Writing better functions[3]

USGS Water Data for the Nation blog

AJ Martinez and ED Hinman | Write functions in R that run better and are easier to understand.

2024

Reproducible Data Science in R: Iterate, don't duplicate[4]

USGS Water Data for the Nation blog

Anthony Martinez | Iterate your R code using the map() function from the purrr package.

I enjoy translating complex data science topics and techniques into accessible articles, ensuring that valuable knowledge is shared widely with audiences of varied technical skill levels.

USGS PUBLICATIONS, DATA, AND SOFTWARE

2025

Water use across the conterminous United States, water years 2010–20[5]

U.S. Geological Survey

L Medalie et al. including **AJ Martinez**. 2025. Water use across the conterminous United States, water years 2010–20, chap. D *of* U.S. Geological Survey Integrated Water Availability Assessment—2010–20. U.S. Geological Survey professional paper. https://doi.org/10.3133/pp1894D.

2025

Status of water-quality conditions in the United States, 2010–20[6]

U.S. Geological Survey

ML Erickson et al. including **AJ Martinez**. 2025.Status of water-quality conditions in the United States, 2010–20, chap. C *of* U.S. Geological Survey Integrated Water Availability Assessment—2010–20. U.S. Geological Survey professional paper. https://doi.org/10.3133/pp1894C.

2025

Water supply in the conterminous United States, Alaska, Hawaii, and Puerto Rico, water years 2010–20[7]

U.S. Geological Survey

GA Gorski et al. including **AJ Martinez**. 2025. Water supply in the conterminous United States, Alaska, Hawaii, and Puerto Rico, water years 2010–20, chap. B *of* U.S. Geological Survey Integrated Water Availability Assessment—2010–20. U.S. Geological Survey professional paper 1894–B. https://doi.org/10.3133/pp1894B.

2025

Local water use and climate drive water stress over the conterminous United States with substantial impacts to fish species of conservation concern[8]

U.S. Geological Survey

EG Stets, OL Miller, MJ Cashman, KA Powlen, **AJ Martinez**, AA Archer, JA Padilla. 2025. *ESS Open Archive*. https://doi.org/10.22541/essoar.173655431.12049152/v1.

2025

Crosswalk table between 12-digit hydrologic unit code (HUC12) and hydrologic region boundaries[9]

U.S. Geological Survey

AJ Martinez, JE Reddy, JA Padilla. 2025. Crosswalk table between 12-digit hydrologic unit code (HUC12) and hydrologic region boundaries. U.S. Geological Survey data release. https://doi.org/10.5066/P13XTCTM.

2025

Monthly ensemble outputs from the National Hydrologic Model Precipitation-Runoff Modeling System and the Weather Research and Forecasting model hydrologic modeling system for the conterminous United States, Alaska, Hawaii, and Puerto Rico for water years[10]

U.S. Geological Survey

AJ Martinez, JA Padilla, GA Gorski. 2025. Monthly ensemble outputs from the National Hydrologic Model Precipitation-Runoff Modeling System and the Weather Research and Forecasting model hydrologic modeling system for the conterminous United States, Alaska, Hawaii, and Puerto Rico for water years. U.S. Geological Survey data release. https://doi.org/10.5066/P1RBMDUT.

2025

Evaluating high, moderate, and low concentrations of groundwater constituents across principal aquifer locations in the conterminous United States[11]

U.S. Geological Survey

E Azadpour, **AJ Martinez**, JA Padilla. Evaluating high, moderate, and low concentrations of groundwater constituents across principal aquifer locations in the conterminous United States. U.S. Geological Survey software release. https://doi.org/10.5066/P13N7XHW.

2025

Analysis of monthly water supply for the United States for water years 2010-2020 using modeled states and fluxes[12]

U.S. Geological Survey

AJ Martinez, HR Corson-Dosch, JA Padilla, GA Gorski. 2025. Analysis of monthly water supply for the United States for water years 2010-2020 using modeled states and fluxes. U.S. Geological Survey software release. https://doi.org/10.5066/P1YTXWKY.

2025

Visualizations of water quality across the conterminous United States using SPARROW 2012 simulated nutrient load estimates[13]

U.S. Geological Survey

AJ Martinez, E Azadpour, JA Padilla, OL Miller, MJ Cashman. 2025. Visualizations of water quality across the conterminous United States using SPARROW 2012 simulated nutrient load estimates. U.S. Geological Survey software release. https://doi.org/10.5066/P14XRQ9R.

2024

Water budget estimation for a water availability assessment across the conterminous United States for water years 2010–2020[14]

U.S. Geological Survey

OL Miller, **AJ Martinez**, MJ Cashman, KA Powlen, JA Padilla, EG Stets. Water budget estimation for a water availability assessment across the conterminous United States for water years 2010–2020. U.S. Geological Survey software release. https://doi.org/10.5066/P14MPRDE.

2024

Formatting and rescaling water use data for the IWAAs National Report[15]

U.S. Geological Survey

AJ Martinez, JR Zemmels, JA Padilla, JL Blodgett. 2024. Formatting and rescaling water use data for the IWAAs National Report. U.S. Geological Survey data release. https://doi.org/10.5066/P1Q6FUKZ.

2024

Water budget results for a water availability assessment across the conterminous United States for water years 2010-2020[16]

U.S. Geological Survey

OL Miller, **AJ Martinez**, MJ Cashman, KA Powlen, JA Padilla,EG Stets. 2024. Water budget results for a water availability assessment across the conterminous United States for water years 2010-2020. U.S. Geological Survey data release. https://doi.org/10.5066/P1CFYEHO.

2024

Intersectional weights between two different 12-digit Hydrologic Unit Code (HUC12) boundaries[17]

U.S. Geological Survey

AJ Martinez, JR Zemmels, JA Padilla. 2024. Intersectional weights between two different 12-digit Hydrologic Unit Code 12(HUC12) boundaries. U.S. Geological Survey data release. https://doi.org/10.5066/P1UANON8.

2020

A guidebook to spatial datasets for conservation planning under climate change in the Pacific Northwest[18]

U.S. Geological Survey

AJ Martinez, AJH Meddens, JM Cartwright. 2020. Chapter 8: Database of unburned areas within fire perimeters. In JM Cartwright (ed.), *A guidebook to spatial datasets for conservation planning under climate change in the Pacific Northwest: Northwest Climate Adaptation Science Center*. https://doi.org/10.5066/P92L1H7O.

NON-USGS PUBLICATIONS

2021

Assessing the quality of fire refugia for wildlife habitat[19]

Forest Ecology and Management

RA Andrus, **AJ Martinez**, GM Jones, AJH Meddens. 2021. Assessing the quality of fire refugia for wildlife habitat. *Forest Ecology and Management* 482, 118868. https://doi.org/10.1016/j.foreco.2020.118868.

2019

Characterizing persistent unburned islands within the Inland Northwest USA[20]

Fire Ecology

AJ Martinez, AJH Meddens, CA Kolden, EK Strand, AT Hudak. 2019. Characterizing persistent unburned islands within the Inland Northwest USA. *Fire Ecology* 15, 1-18. https://doi.org/10.1186/s42408-019-0036-x.

2019 **The potential importance of unburned islands as refugia for the persistence of wildlife species in fire-prone ecosystems**[21]

Ecology and Evolution

J Steenvoorden, AJH Meddens, **AJ Martinez**, LJ Foster, WD Kissling. 2019. The potential importance of unburned islands as refugia for the persistence of wildlife species in fire-prone ecosystems. *Ecology and Evolution* 9 (15), 8800-8812. https://doi.org/10.1002/ece3.5432.

2019 **An Assessment of Fire Refugia Importance Criteria Ranked by Land Managers**[22]

Fire

AJ Martinez, AJH Meddens, CA Kolden, AT Hudak. 2019. An Assessment of Fire Refugia Importance Criteria Ranked by Land Managers. *Fire* 2 (2), 27. https://doi.org/10.3390/fire2020027.

PRESENTATIONS (PRESENTER)

2020 **Forest Measurements in Silviculture**

Washington State University — Pullman, WA

AJ Martinez
Guest lecture

2018 **High-resolution Lidar from Unmanned Aerial Vehicles for Forestry Applications**

University of Idaho GIS Day — Moscow, ID

AJ Martinez, AJH Meddens, CA Silva, LA Vierling L, JUH Eitel
Poster presentation.

2017 **Spatiotemporal patterns of unburned areas within fire perimeters in the northwestern United States from 1984 to 2014**

Association for Fire Ecology International Fire Congress — Orlando, FL

AJ Martinez, AJH Meddens, CA Kolden, JA Lutz, JT Abatzoglou, AT Hudak
Oral presentation

2017 **Detection and Prediction of Wildfire Tree Mortality Using Landsat and lidar**

University of Washington Capstone Poser Event — Seattle, WA

AJ Martinez
Poster Presentation

LINKS

1. https://alliance-uidaho.primo.exlibrisgroup.com/permalink/01ALLIANCE_UID/2vsn7u/alma996340112901851
2. https://waterdata.usgs.gov/blog/rds-functions-that-work-for-you/

3. https://waterdata.usgs.gov/blog/rds-better-functions/
4. https://waterdata.usgs.gov/blog/rds-iterate/
5. https://doi.org/10.3133/pp1894D
6. https://doi.org/10.3133/pp1894C
7. https://doi.org/10.3133/pp1894B
8. https://doi.org/10.22541/essoar.173655431.12049152/v1
9. https://doi.org/10.5066/P13XTCTM
10. https://doi.org/10.5066/P1RBMDUT
11. https://doi.org/10.5066/P13N7XHW
12. https://doi.org/10.5066/P1YTXWKY
13. https://doi.org/10.5066/P14XRQ9R
14. https://doi.org/10.5066/P14MPRDE
15. https://doi.org/10.5066/P1Q6FUKZ
16. https://doi.org/10.5066/P1CFYEHO
17. https://doi.org/10.5066/P1UANON8
18. https://doi.org/10.5066/P92L1H7O
19. https://doi.org/10.1016/j.foreco.2020.118868
20. https://doi.org/10.1186/s42408-019-0036-x
21. https://doi.org/10.1002/ece3.5432
22. https://doi.org/10.3390/fire2020027